Concentration of Random Sums: A Short Sample Paper

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Abstract

We give a short proof that a sum of n bounded independent variables stays close to its mean. The bound in Theorem 2 is tight up to a constant, see Figure 1 and Table 1. This text is a sample for the pdfview viewer.

1 Introduction

Let $X_1, \dots, X_n$ be independent variables with $a_i \le X_i \le b_i$, and let $S_n = \sum_{i=1}^{n} X_i$. Classical results [1, 2] bound the probability that S_n is far from its mean. We show that the bound is tight for all n, and we give the constant in Section 3.[1] The main tool is the moment generating function

$$M(\lambda) = \mathbb{E}\, e^{\lambda (S_n - \mathbb{E}S_n)}, \qquad \lambda \in \mathbb{R}. \tag{1}$$

2 The main result

Lemma 1. *If $a \le X \le b$ and $\mathbb{E}X = 0$, then $\mathbb{E}\, e^{\lambda X} \le \exp\bigl(\lambda^2 (b-a)^2/8\bigr)$.*

Proof. The function $x \mapsto e^{\lambda x}$ is convex, so it lies under its chord on $[a, b]$. Take the expectation and bound the logarithm of the result by its second-order Taylor polynomial. □

Theorem 2. *For every $t > 0$,*

$$\Pr\bigl(|S_n - \mathbb{E}S_n| \ge t\bigr) \le 2 \exp\!\left(-\frac{2t^2}{\sum_{i=1}^{n} (b_i - a_i)^2}\right). \tag{2}$$

Proof. By Markov's inequality and (1), $\Pr(S_n - \mathbb{E}S_n \ge t) \le e^{-\lambda t} M(\lambda)$. Lemma 1 bounds each factor of $M(\lambda)$. Choose $\lambda = 4t / \sum_i (b_i - a_i)^2$, and use the same argument for $-S_n$. □

3 Results

Figure 1 compares the bound (2) with the exact tail for fair coins. Table 1 gives the values for some n. The constant c depends on the dimension d of the space.

[1] A longer version of this text has the full proofs.

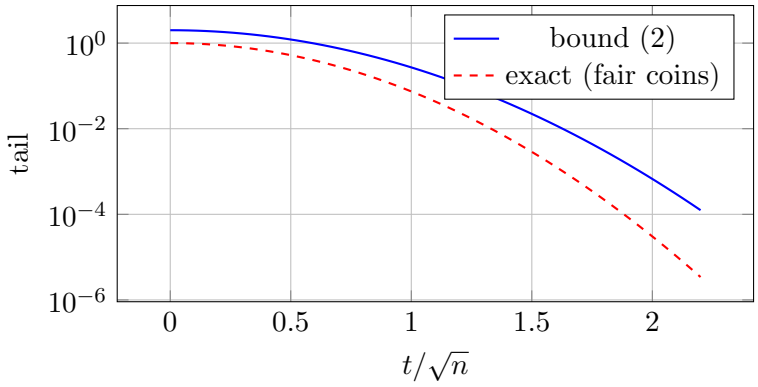


Figure 1: The bound and the exact tail.

n	t	bound	exact
10	3	0.330	0.109
100	10	0.271	0.057
1000	40	0.081	0.012

Table 1: Some values of the bound and of the exact tail.

4 Discussion

The bound of Theorem 2 needs only the range of each variable. A bound with the variance is sharper when the variance is small; see [2]. The code for the figures is at `https://example.com/concentration`. We end with an open question about the case of dependent variables.

References

[1] W. Hoeffding. Probability inequalities for sums of bounded random variables. *J. Amer. Statist. Assoc.*, 58:13–30, 1963.

[2] S. Boucheron, G. Lugosi and P. Massart. *Concentration Inequalities*. Oxford University Press, 2013.